

Isolation of peptide hormones with pleiotropic activities in the freshwater crab, *Oziotelphusa senex senex*

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Abstract

In the present study, the sinus gland peptide hormones were isolated from the freshwater field crab, *Oziotelphusa senex senex* by reverse-phase high performance liquid chromatography (HPLC) and the dual biological activity of isolated peptides were determined. Of the six eluted peaks, Oss-SGP 2 and 3 exhibited significant mandibular organ-inhibiting activity. In addition to mandibular organ-inhibiting activity, Oss-SGP 2 exhibited potent Y-organ inhibiting activity in vitro and Oss-SGP 3 exhibited significant hyperglycemic activity in vivo. Peptide fractions Oss-SGP 5 and 6 also exhibited significant hyperglycemic activity. The present study not only demonstrated the presence of pleiomorphic forms of peptides but also pleiotropic activities of eyestalk peptides of the crab *O. senex senex*.

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1. Introduction

Oziotelphusa senex senex is a freshwater edible crab normally inhabiting in rice fields and irrigation canals throughout South India. Although originally restricted to freshwater, they can also survive in 100% seawater. Recently, outbreak of white spot disease and susceptibility of prawn species to this disease resulted in search for alternate species for aquaculture. In view of a rapid growth rate, high meat content, excellent palatability and resistance to pathogens has led to the development of a burgeoning aquaculture industry for several crab species in South India. Although research

into *Oziotelphusa* has concentrated on aspects pertaining to reproduction (Reddy et al., 2004a,b) and metabolism (Reddy and Ramamurthi, 1982; Reddy, 1990), a paucity of biological data has hampered culture attempts of this species. The aim of this study was to describe the purification of eyestalk peptide hormones and to evaluate their physiological activities in *O. senex senex*.

In crustaceans, the neurohemal tissue, X-organ-sinus gland complex is located in the eyestalks, which synthesizes and releases the neuroendocrine peptide hormones collectively called crustacean hyperglycemic hormone-family (CHH-family) neuropeptides (Keller, 1992; Reddy and Ramamurthy, 1998; Lacombe et al., 1999; Huberman, 2000; Diwan, 2005; Chen et al., 2005). The CHH-family neuropeptide hormones include hyperglycemic hormone (CHH), molt-inhibiting

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hormone (MIH), vitellogenesis-inhibiting hormone (VIH) and mandibular organ-inhibiting hormone (MOIH). The members of CHH-family peptides share the following characteristics: (1) consist of 72–78 amino acid residues and (2) have similar amino acid sequences with six cysteines, within the molecule being conserved (Keller, 1992; Reddy and Ramamurthi, 1999; Lacombe et al., 1999; Chen et al., 2005). Functionally CHH regulates carbohydrate metabolism (Santos and Keller, 1993). MIH inhibits the synthesis and release of ecdysteroid from the Y-organs (Aiken and Waddy, 1992; Spaziani et al., 1994; Lee et al., 1998; Chung and Webster, 2003). The VIH regulates the process of vitellogenesis negatively, where it inhibits the synthesis of vitellogenin (Soyez et al., 1987, 1991; Aguilar et al., 1992; Huberman et al., 1995). The MOIH inhibits the synthesis and release of methyl farnesoate (MF) by mandibular organ (MO) (Yudin et al., 1980; Tamone et al., 1997; Reddy and Ramamurthi, 1999; Kalavathy et al., 1999; Laufer and Biggers, 2001; Nagaraju et al., 2004; Reddy et al., 2004b).

In addition to structural similarities, overlapping physiological effects have also been observed in this peptide family, viz., a peptide with both molt-inhibiting and hyperglycemic activities has been isolated from *Homarus americanus* (Chang et al., 1990), *Penaeus vannamei* (Sefiani et al., 1996) and *Scylla serrata* (Reddy and Reddy, 2006). Yasuda et al. (1994) reported one of the two variants of hyperglycemic hormone isolated from *Procambarus clarkii* exhibited more potent molt-inhibiting activity. On the other hand, another peptide which exhibits more potent molt-inhibiting hormone activity than molt-inhibiting activity exhibited by hyperglycemic hormone has been isolated from *Carcinus maenas*, *P. clarkii*, *Orconectes limosus* and *Penaeus japonicus* (Webster, 1991; Webster and Keller, 1986; Yasuda et al., 1994; von Gliscynski, 1994; Yang et al., 1996). The presence of several isoforms of the peptide hormones were also demonstrated in many crustacean species including the crabs *Cancer pagurus* (Wainwright et al., 1996), *C. maenas* (Dirksen et al., 2001), *O. senex senex* (Nagaraju et al., 2005) and *S. serrata* (Reddy and Reddy, 2006), the prawns *Penaeus semisulcatus* (Khayat et al., 1998) and *P. vannamei* (Sefiani et al., 1996), the crayfishes *Procambarus bouvieri* (Aguilar et al., 1996) and *P. clarkii* (Yasuda et al., 1994) and the lobster *H. americanus* (Chang et al., 1990; De Kleijn et al., 1998). Though extensive work has been done on the dual biological activity of CHH-family peptide hormones, all the earlier studies were done using single peptide peak isolated from the eyestalk nervous tissue.

In the present study we report the physiological activities of all the peptides isolated from eyestalk nervous tissue of the crab *O. senex senex*.

2. Materials and methods

2.1. Procurement of animals and housing

Intact, intermolt (stage C₄) adult freshwater crabs, *O. senex senex* Fabricius with a body weight of 30±3 g (carapace width of 39±3 mm) were collected from the rice fields and irrigation canals around Tirupati (13°36' N, 79°21'E), Andhra Pradesh, India. Animals were housed 6–8 per glass aquaria (length/width/height=60:30:30 cm) with sufficient ambient medium (salinity: 0.5 ppt) and transferred to fresh medium everyday. They were acclimatized to the laboratory conditions (temperature 27±1 °C; relative humidity 75% and a light period of 12 h) for 7 days. During their sojourn, the crabs were fed with sheep meat ad libitum. Feeding was stopped 1 day before the commencement of the experiment.

2.2. Separation of sinus gland peptides

The sinus gland peptides were separated by reverse-phase HPLC as previously described (Wainwright et al., 1996). In brief, eyestalks were removed from ice-anesthetized intermolt (stage C₄) crabs. Sinus glands were dissected from eyestalks and stored at –80 °C for further use. One hundred sinus glands were homogenized in 2 M acetic acid and then centrifuged at 14,000×g for 5 min. This sinus gland extract was further fractionated using μ -Bondapak Phenyl column (3.9×300 mm; Waters, Milford, MA), with a linear gradient of 30–80% acetonitrile/water containing 0.1% trifluoroacetic acid. The eluted fractions were monitored at 210 nm and the individual peaks were collected manually and designated as Oss-SGP 1, 2, 3, 4, 5 and 6. These peptide fractions were stored at 4 °C after drying under vacuum.

2.3. Bioassay for mandibular organ-inhibiting hormone (MOIH) activity

The mandibular organ-inhibiting hormone activity of each peptide was measured by using in vitro culture of mandibular organs of crab *O. senex senex* by measuring methyl farnesoate synthesized by mandibular organ. Mandibular organs were dissected in pairs from ice-anesthetized, intermolt (stage C₄), male crabs and washed with ice-cold M 199 medium. Mandibular organs were subsequently cultured pairwise (control mandibular organ

versus treated contralateral mandibular organ) in micro-titer plates with 200 μ l of the same medium with or without sinus gland peptides (two sinus gland equivalents). Following incubation, mandibular organ was homogenized in 1000 μ l of CH_3CN , 4% NaCl (W/V) and centrifuged at 1000 $\times g$ for 10 min at 4 °C. Methyl farnesoate was partitioned thrice against 500 μ l of hexane.

Methyl farnesoate was measured using the method described by Borst and Tsukimura (1991) with minor modifications. The extracted methyl farnesoate was separated using reverse-phase HPLC (μ -Bondapak Phenyl column, 2.9 $\times$ 300 mm; Waters, Milford, MA), with an isocratic elution using 70% acetonitrile in water (1.0 ml/min). Methyl farnesoate peak was detected at 254 nm and authenticated using standard methyl farnesoate (generously supplied by Prof. R. Ramamurthi, INSA Senior Scientist, Sri Venkateswara University, Tirupati).

2.4. Bioassay for molt-inhibiting hormone (MIH) activity

The molt inhibiting hormone activity of each peptide was measured by using in vitro culture of Y-organs of the crab and an ecdysteroid radioimmunoassay (RIA). Y-organs were dissected away from the intermolt (stage C_4) crabs and washed for 30 min in crab ringer solution (Van Harreveld, 1936) at 4 °C. Individual Y-organs were incubated in 24-well tissue culture plates in 1.0 ml of M 199 medium with 2 μ l of non-sinus gland extract or 2 μ l of sinus gland peptide (equivalent to 2 sinus glands). Chelate leg muscle homogenate (1% W/V) was prepared in 2 M acetic acid and centrifuged at 14,000 $\times g$ for 5 min. The collected supernatant was dried under vacuum and was used as non-sinus gland extract (equivalent to 200 μ g tissue). All incubations were carried out at 16 °C. After 24 and 48 h, 50 μ l of the culture medium was removed from each well and prepared for ecdysteroid assay. At the end of the experiment, the Y-organ was blotted and weighed on a microbalance.

Ecdysteroids were extracted from the Y-organ incubation medium into 75% methanol and the RIA was conducted as previously described (Soumoff and O'Connor, 1982). Ecdysteroid concentrations were normalized for Y-organ weight. Ecdysteroid antibodies were generously supplied by Dr. W. E. Bollenbacher, University of North Carolina, Chapel Hill, USA.

2.5. Bioassay for hyperglycemic hormone

The bioassay of hyperglycemic activity was done essentially according to the method reported earlier

(Leuven et al., 1982). Eyestalk-ablated (24 h) intermolt (stage C_4) male crabs (body weight 30 ± 2 g) were used for crustacean hyperglycemic hormone assay. All the sinus gland peptide fractions (2 sinus gland equivalents) in 10 μ l crustacean saline were injected into crabs which had been eyestalk ablated 1 day before. Control crabs received 20 μ l of physiological saline. Injections were made through the arthrodial membrane at the base of the 3rd pair of walking legs. Hemolymph (50 μ l) was collected prior to injection and 2 h after injection. Glucose in each hemolymph sample was measured by the glucose enzyme reagent (Sigma). Maximum hyperglycemic response was observed 2 h after injection of eyestalk extract in the crab, *O. senex senex* (Reddy and Ramamurthi, 1982) hence 2 h time point was selected for hyperglycemic assay.

2.6. Statistical analysis

Results obtained were subjected to the following statistical analyses. The mean, standard deviation (S.D.) and Analysis of Variance (ANOVA) were made using SPSS 13.0 version (SPSS Inc., Chicago, IL).

3. Results

3.1. Effect of eyestalk ablation on MF synthesis

Bilateral eyestalk ablation (24 h) stimulated methyl farnesoate synthesis in the mandibular organs of the crab *O. senex senex* (Fig. 1). The increase in the synthesis is 58.38% higher than the methyl farnesoate synthesis by mandibular organ isolated from intact crabs. The

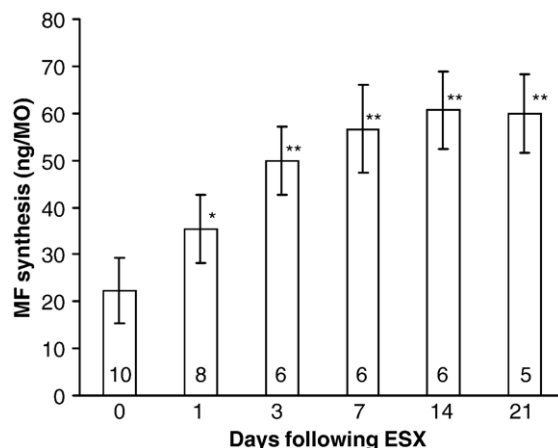


Fig. 1. The effect of eyestalk ablation (ESX) on methyl farnesoate secretion by the mandibular organ in the crab *Oziotelphusa senex senex*. Vertical bars represent mean $\pm$ S.D. Number in the bars represent sample size. * $p < 0.001$, ** $p < 0.0001$.

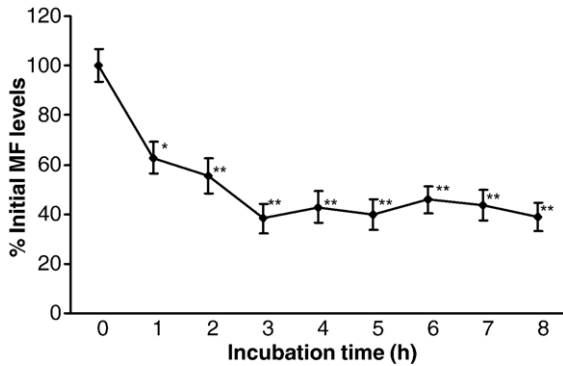


Fig. 2. The time–response of sinus gland extract on methyl farnesoate secretion by the mandibular organ. Values are mean $\pm$ S.D. of six individual observations. Methyl farnesoate levels after incubation were expressed as % of the initial methyl farnesoate levels. Methyl farnesoate levels were measured at the indicated time points after incubation of mandibular organ with extract of two sinus gland equivalents. * $p < 0.001$, ** $p < 0.0001$.

stimulation of methyl farnesoate synthesis continued up to 7 days and after 7 days no further increase in the methyl farnesoate synthesis was observed.

Incubation of mandibular organs with sinus gland extract (one sinus gland equivalent) caused significant ($p < 0.001$; 37.19%) inhibition in methyl farnesoate synthesis within 1 h and reached a maximum (61.61%) response after 3 h (Fig. 2). These studies support the presence of an MOIH within the eyestalks of *O. senex senex*. In the subsequent experiments, methyl farnesoate was estimated 3 h after treatment.

Dose–response studies (Fig. 3) show significant ($p < 0.0001$) correlation between the inhibition of methyl farnesoate secretion and the amount of sinus gland equivalents incubated for the values ranging from 0 to

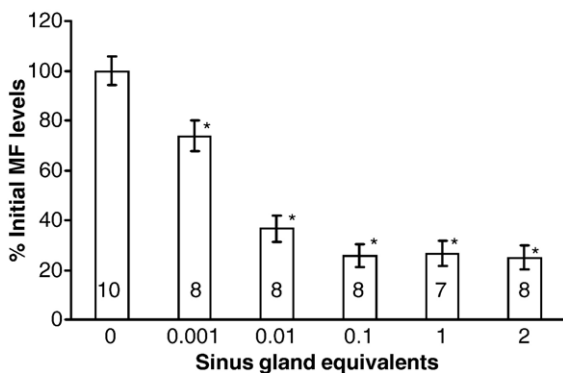


Fig. 3. Dose–response of sinus gland (SG) extract on methyl farnesoate synthesis in the mandibular organ. Vertical bars represent mean $\pm$ S.D. Number in the bars represent sample size. Methyl farnesoate levels were determined in the mandibular organs after 3 h of incubation with sinus gland extract and expressed as % of the initial MF levels. * $p < 0.001$.

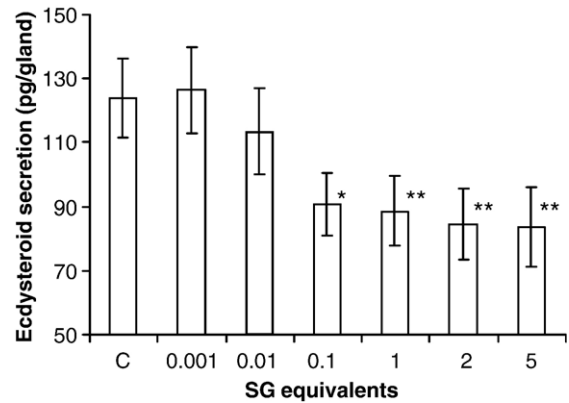


Fig. 4. Dose–response of sinus gland (SG) extract on ecdysteroid secretion by Y-organs in vitro of the crab *Oziotephusa senex senex*. Vertical bars represent mean $\pm$ S.D. of six individual observations. Ecdysteroid levels were determined in the medium after incubating Y-organs with or without sinus gland extract for 24 h. * $p < 0.001$, ** $p < 0.0001$.

0.1 sinus gland equivalents. The latter dose giving the maximum effect, but even the dose of 0.001 sinus gland equivalent caused significant ($p < 0.001$; 26.32%) inhibition in methyl farnesoate synthesis.

3.2. Effect of eyestalk extract on ecdysteroid secretion in vitro

The ecdysteroid secretion decreased in the Y-organs incubated in vitro with sinus gland extract of crab *O. senex senex* (Fig. 4). The decrease in levels of ecdysteroid was dose dependent. Maximum inhibition

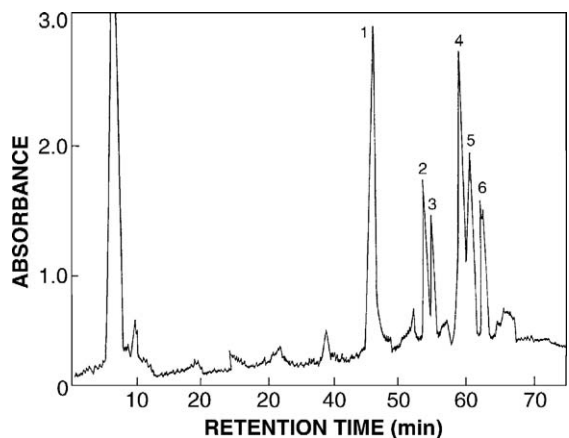


Fig. 5. Reverse-phase chromatogram of peptides contained within the sinus glands of *Oziotephusa senex senex*. The extract of 100 sinus glands was separated by RP-HPLC and the chromatograph conditions were; column, Waters μ -Bondapak phenyl (2.9 $\times$ 300 mm); solvent A, 0.11% trifluoroacetic acid; solvent B, 60% acetonitrile containing 0.1% trifluoroacetic acid; gradient 30–80% solvent B over 75 min, 1 ml/min.

(33.57%) in ecdysteroid synthesis occurred at two sinus gland equivalents, but even at the dose of 0.1 sinus gland equivalent caused significant ($p < 0.001$; 26.65%) inhibition in ecdysteroid synthesis.

3.3. Isolation and characterization of eyestalk peptides

A chromatogram obtained by reverse-phase HPLC with the extracts from 100 sinus glands of the crab *O. senex senex* is presented in Fig. 5. The chromatogram shows the presence of a number of peaks with the majority eluting after 40 min. Fractionation of sinus gland extract resulted in elution of six major peptide peaks and designated as Oss-SGP 1, 2, 3, 4, 5 and 6.

3.4. Biological activities of the different peptides

HPLC-separated peptides were used at two sinus gland equivalents to study their effect(s) on regulation of methyl farnesoate synthesis by mandibular organs, ecdysteroid secretion by Y-organs in vitro and hemolymph glucose level in vivo (Tables 1–3). Among the six peptides, two peptides (Oss-SGP 2 and 3) exhibited significant mandibular organ inhibiting activity (55.41% and 72.64%, respectively), with a maximal inhibition for Oss-SGP 3 (Table 1). Incubation of MO with Oss-SGP 1, 4, 5 and 6 showed almost no activity in altering methyl farnesoate synthesis by the mandibular organs.

Ecdysteroid secretion from the Y-organ in vitro after incubating with peptide fractions (Oss-SGP 1–4) is shown in Table 2. It is clear from the data that Oss-SGP 2 is very potent in inhibiting Y-organ to secrete ecdysteroid. Maximum inhibition in ecdysteroid secretion was observed 48 h (59.12%) of incubation with

Table 2

The effect of sinus gland extract and sinus gland derived peptides on ecdysteroid secretion from the Y-organs of *Oziotelphusa senex senex*

Sample	Ecdysteroid secretion (pg/gland)	
	24 h	48 h
Control (non-sinus gland extract)	125.83 ± 12.32	128.88 ± 15.49
Sinus gland extract	85.43 ± 11.87* (–32.10)	70.84 ± 14.55* (–45.03)
Peptide 1	122.44 ± 11.66 ^{NS} (–2.77)	126.84 ± 10.68 ^{NS} (–1.58)
Peptide 2	71.33 ± 8.63* (–43.31)	52.68 ± 6.66* (–59.12)
Peptide 3	132.78 ± 12.80 ^{NS} (5.52)	135.66 ± 19.88 ^{NS} (5.26)
Peptide 4	120.66 ± 12.44 ^{NS} (–4.11)	123.66 ± 11.79 ^{NS} (–4.05)
Peptide 5	128.63 ± 11.84 ^{NS} (–2.23)	127.52 ± 13.41 ^{NS} (–1.06)
Peptide 6	126.09 ± 11.89 ^{NS} (0.21)	129.37 ± 12.56 ^{NS} (0.38)

Y-organs were incubated with non-sinus gland extract (1% W/V) or two sinus gland equivalents of extract or peptide fraction for 24 and 48 h at 16 °C. Ecdysteroids in culture medium was measured by radioimmunoassay (RIA). Each value represents mean ± S.D. of six samples. Values in parentheses are % change from control. Asterisk denotes significance at $p < 0.001$. NS = not significant.

peptide 2. Whereas Oss-SGP 1, 3–6 did not cause any change in ecdysteroid secretion from Y-organs.

The in vivo effect of these sinus gland peptides on hemolymph glucose regulation was also investigated. Eyestalk ablated (24 h) male crabs were used for this study. Hemolymph samples were collected from each crab before and 2 h after injection. Among all peptides, injection of Oss-SGP 3, 5 and 6 induce a hyperglycemic response with a maximal activity for Oss-SGP 6 (148.89%) and Oss-SGP 3 (140.00%). Injection of Oss-SGP 1, 2 and 4 and saline do not significantly increase glycemia (Table 3).

Table 3

The effect of injection of sinus gland extract and sinus gland derived peptides on hemolymph glucose level in the crab, *Oziotelphusa senex senex*

Sample	Glucose levels (mg/ml)	
	Before injection	120 min after injection
Control (saline injected)	0.43 ± 0.07	0.45 ± 0.06 (4.65)
Sinus gland extract	0.44 ± 0.11	0.98 ± 0.27* (122.72)
Peptide 1	0.41 ± 0.12	0.44 ± 0.11 ^{NS} (7.32)
Peptide 2	0.44 ± 0.11	0.43 ± 0.08 (–2.27)
Peptide 3	0.45 ± 0.12	1.08 ± 0.21* (140.00)
Peptide 4	0.42 ± 0.13	0.44 ± 0.15 ^{NS} (4.76)
Peptide 5	0.46 ± 0.14	0.96 ± 0.22* (108.70)
Peptide 6	0.45 ± 0.11	1.12 ± 0.24* (148.89)

Two sinus gland equivalents were injected into assay crabs. Hemolymph glucose levels was measured both immediately before and 120 min after injection of sample. Control crabs were injected with 10 µl of saline. Values are mean ± S.D. of 8 crabs. Values in parentheses are % change from before injection samples. * $p < 0.0001$; NS = not significant.

Table 1

The effect of sinus gland extract and sinus gland derived peptides from the crab, *Oziotelphusa senex senex* on methyl farnesoate secretion by mandibular organ (MO) in vitro

Sample	MF levels (ng/MO)
Control (saline injected)	25.37 ± 6.91
Sinus gland extract	12.38 ± 2.39* (–51.20)
Peptide 1	26.64 ± 4.31 ^{NS} (5.0)
Peptide 2	11.31 ± 3.14* (–55.41)
Peptide 3	6.94 ± 1.17* (–72.64)
Peptide 4	25.31 ± 3.18 ^{NS} (–0.23)
Peptide 5	26.67 ± 4.91 ^{NS} (5.12)
Peptide 6	26.01 ± 3.48 ^{NS} (2.52)

Methyl farnesoate levels were measured after 3 h incubation of mandibular organ with or without sinus gland extract or sinus gland peptide fraction (two sinus gland equivalents). Values are mean ± S.D. of 8 samples. Values in parentheses are % change from control. Asterisk denotes significance at $p < 0.001$; NS = not significant.

4. Discussion

The present investigation provides evidence for an inhibitory effect of eyestalk peptides on ecdysteroid secretion from Y-organs and methyl farnesoate synthesis by mandibular organs in *Oziotelphusa*. Eyestalk ablation removes the source of inhibition of not only the Y-organ but also the MO (Borst et al., 1994; Wainwright et al., 1996; Liu et al., 1997; Nagaraju et al., 2004). In fact, eyestalk removal has been shown to increase hemolymph MF titers in *Libinia emarginata* (Laufer et al., 1987), *H. americanus* and *Orconectes virilis* (Tsukimura et al., 1989). Nagaraju et al. (2004) have reported that the weight of the mandibular organs increased significantly after eyestalk ablation in the crab *O. senex senex*. It is clear that sinus gland peptides inhibit the Y-organs to secrete ecdysteroids and mandibular organs to secrete methyl farnesoate. This is consistent with earlier studies that eyestalk possesses both molt-inhibiting hormone and mandibular organ-inhibiting hormone (see reviews Reddy and Ramamurthi, 1999; Diwan, 2005; Chen et al., 2005).

The reverse-phase HPLC profile of the crab *O. senex senex* sinus gland peptides is similar to that reported for other crustacean sinus gland peptides with respect to the retention time. Multiple peak fractions with several physiological activities have also been demonstrated in other decapods such as the spider crab, *L. emarginata* (Liu et al., 1997), the edible crab, *C. pagurus* (Wainwright et al., 1996), the crayfishes *P. bouvieri*, *O. limosus* and *P. clarkii* (Huberman and Aguilar, 1986; Kegel et al., 1991; Yasuda et al., 1994), the shrimp *Palaemon serratus* (van Wormhoudt et al., 1984) and the lobster *H. americanus* (Soyez et al., 1994).

From the data it is evident that several peptides have been found to have dual biological effects. Oss-SGP 2 and 3 induce a significant mandibular organ-inhibiting activity and Oss-SGS 2 also possesses a significant ecdysteroid inhibiting activity. In assays using ablated crabs, three peptides (Oss-SGP 3, 5 and 6) induce a significant hyperglycemia. It has been shown that the MOIH purified from the sinus gland of the spider crab *L. emarginata* inhibited methyl farnesoate synthesis in vitro and exhibited hyperglycemic activity when injected into crabs (Liu et al., 1997). It has also been shown that hyperglycemic hormone isolated from *C. pagurus* inhibited methyl farnesoate synthesis by the mandibular organs in vivo (Wainwright et al., 1998). Earlier Chang et al. (1990) from *H. americanus*, Sefiani et al. (1996) from *P. vannamei*, and Reddy and Reddy (2006) from *O. senex senex* have characterized a neuropeptide that displayed molt-inhibiting activity

and also involved in the regulation of hemolymph sugar level. The present data is also in agreement with the presence of different isoforms of peptide hormone demonstrated in other decapods such as crayfish *P. bouvieri*, *P. clarkii* and *O. limosus* (Huberman and Aguilar, 1988; Yasuda et al., 1994; Ollivaux and Soyez, 2000), the shrimp *P. serratus* (van Wormhoudt et al., 1984), the lobster, *H. americanus* (Tensen et al., 1991; Soyez et al., 1994) and the penaeid shrimps, *P. japonicus* (Yang et al., 1995, 1997).

Both the mandibular organ inhibiting peptides identified in the present study possess some other biological activity. The dual biological functions of these peptides may be due to high amino acid sequence similarity with other peptide hormones and is consistent with the observations in the shore crab, *C. maenas* (Webster, 1991). CHH receptors have been identified in the Y-organ membrane preparations suggesting that the role of hyperglycemic hormone in the regulation of ecdysteroid synthesis (Webster and Keller, 1986; Webster, 1993; Chung and Webster, 2003). Webster (1998) also has shown that both MIH and MOIH are co-localized in the X-organ though they are functionally distinct. Lu et al. (2000) have suggested that these peptides resulted from duplication of ancestral hyperglycemic hormone gene, with subsequent divergence producing the functional molt-inhibiting hormone and mandibular organ-inhibiting hormones still possessing the overlapping biological activity. They also found that the overall structure of MIH gene is very similar to MOIH and possesses 67% similarity in the coding regions of gene. Based on the identity between these two peptide hormones, Lu et al. (2000) concluded that the genes are clustered and diverged following a gene duplication event that occurred much further in the past than the emergence of MOIH isoforms. With the passage of time, they have evolved to carry out different roles.

To our knowledge, our data represent the comprehensive biological functions of all peptides isolated from sinus glands of a decapod crustacean. From the results, it is clear that at least two different peptides are responsible for mandibular organ inhibition in *Oziotelphusa* that caused significant inhibition of methyl farnesoate synthesis from the mandibular organs. It is also clear that among these, one peptide exhibited significant inhibition on ecdysteroid synthesis from Y-organ and the other possesses significant hyperglycemic activity. Besides, the sinus gland also possesses two more hyperglycemic hormones. The designation of these peptides as molt-inhibiting hormone, mandibular organ-inhibiting hormone and hyperglycemic hormone

is a consequence of the observed physiological activity. Because these peptides possess at least some other biological activity, the designation of these peptides as the molt-inhibiting hormone, the mandibular organ-inhibiting hormone and the hyperglycemic hormone is premature at this point of time.

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